

Niranga Nayanajith

Undergraduate Student | Cybersecurity Enthusiast |
Software Developer

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About Me

I am an undergraduate student with a deep passion for cybersecurity, artificial intelligence, and software development. I enjoy solving problems, learning new technologies, and building practical solutions. I continuously explore areas such as full-stack development, backend/frontend engineering, and Windows application development, while strengthening my core programming and security knowledge.

EDUCATION

Bachelor of Science in Computing and Information Systems
Sabaragamuwa University of Sri Lanka

- Expected Graduation: December 2028

SKILLS

- Software development expertise
- Programming languages: Python, TypeScript, JavaScript, Ballerina, C#, Java
- Web frameworks: Angular, Next.js, Node.js
- Backend services and APIs
- Systems monitoring and observability
- Distributed systems architecture
- Containerization fundamentals
- Linux command-line proficiency

- Version control with Git/GitHub
- Networking and security fundamentals
- Security principles: CIA triad and risk assessment
- Network scanning tools: Nmap
- Application packaging with PyInstaller and experience with Mamba
- Technical documentation with LaTeX and Jupyter notebooks

CERTIFICATIONS & LICENSES

- Fundamentals of Game Hacking Development – Red Team Leaders
- Cybersecurity Fundamentals – IBM SkillsBuild Student Ambassador Program
- Data Science for Cybersecurity – Red Team Leaders
- Build Your Own AI Chatbot with Microsoft Foundry – Microsoft Learn Student Ambassadors (Sri Lanka)
- AI Buildathon – IEEE Computer Society of KDU
- Networking Basics – Cisco Networking Academy
- SQL Injection – Simplilearn
- Ethical Hacking: Mobile Platforms and Network Architecture – Great Learning
- Introduction to Cyber Security – Great Learning
- Responsive Web Design – freeCodeCamp
- Introduction to Programming Using Python – Kaggle
- Basic in Python – HackerRank
- C Language Gold Level – HackerRank

PROJECTS

1. NailVirtuoso (AI Marketplace & Supreme Admin Console)

Technologies: Next.js, TypeScript, Firebase, Gemini API

- Engineered a “Supreme Admin Console” with real-time API health diagnostics, status dashboards, and Gemini quota monitoring (tracking 1,500 requests/day).
- Implemented a robust Role-Based Access Control (RBAC) system with “Divine Guard” route protection and secure admin-only API endpoints.
- Leveraged Firestore onSnapshot() for live data synchronization across system statistics, inventory, and pending queues.

2. Webify.me (AI-Driven Full-Stack Automation)

Technologies: React, Python (Flask), MongoDB, Gemini API

- Built a complex backend that transforms raw topics into structured content (PPTX), managing communication between a React frontend and a Python-based processing engine.
- Handled large-scale integrations with external APIs (Gemini, Unsplash) and implemented asynchronous task management for content generation

3. 2N Share (Cross-Platform Networking Utility)

Technologies: Python, Flask, Socket, Threading, PyInstaller

- Engineered a multi-threaded backend enabling localized file-sharing with automated local IP discovery for zero-config networking.
- Implemented secure filename handling to prevent directory traversal attacks and ensured data integrity during high-speed transfers.
- Utilized PyInstaller to package optimized, standalone executables for cross-platform reliability.

4. NailDetectAI (Deep Learning Virtual Segmentation)

Technologies: PyTorch, U-Net, Flask, Docker, Gunicorn

- Packaged a deep learning model using Docker and Gunicorn for reproducible and scalable deployment.
- Implemented image segmentation (U-Net) for virtual visualization, demonstrating advanced algorithmic and mathematical problem-solving skills.

5. Voice Assistant & Virtual Mouse (HCI & OS Automation)

Technologies: OpenCV, MediaPipe, PyAutoGUI, Windows APIs

- Solved hardware-software lag using weighted moving average algorithms for smooth real-time interaction with low CPU overhead.
- Integrated Python scripts with low-level Windows APIs for direct system control, showcasing deep OS-level understanding.

6. High-Performance PDF Difference Engine

Technologies: Python, C++, pybind11, Tesseract OCR, pypdfium2

- Engineered a stateless comparison engine utilizing a multi-stage architecture to solve memory exhaustion and layout shift issues in large-scale PDF processing.
- Developed a custom "Kv spatial mechanism" in C++ via pybind11, upgrading standard LCS algorithms with Jaccard similarity thresholds to surgically eliminate false positives.
- Architected a memory-safe 10-page batching system with a `find_safe_seam()` algorithm to maintain sentence integrity during high-volume document analysis.
- Implemented a Hybrid Extraction Pipeline using pdfplumber and Tesseract OCR with image-preprocessing (2.0x contrast boost) for pixel-perfect coordinate mapping.

7. Wikizero AI (Custom Chatbot Marketplace)

Technologies: Next.js, TypeScript, Google Gemini API, Firebase

- Engineered a dynamic platform enabling users to generate, configure, and share custom AI personas powered by the Gemini API.
- Implemented an SEO-optimized architecture featuring dynamic routing and auto-generated sitemaps to ensure indexed discoverability of user-generated bots.
- Developed a robust backend for real-time chat log persistence and profile management using Firestore and TypeScript for end-to-end type safety.

8. ðUnicode (System-Wide Phonetic Converter)

Technologies: C#, .NET Framework, Windows APIs (User32.dll)

- Developed a lightweight Windows utility to bridge the gap between legacy font mappings and modern Unicode using phonetic (Singlish) input.
- Implemented a global keyboard hook and clipboard integration, enabling real-time conversion across all system applications (Office, Browsers, IDEs).
- Built an optimized Regex-based conversion engine to handle complex character ligatures and legacy font transitions with zero latency.

9. SmileHub (Patient Management & Scheduling)

Technologies: Next.js, Tailwind CSS, Firebase

- Developed a full-stack dental healthcare portal featuring an automated appointment booking system to streamline clinic operations and patient intake.
- Designed a high-performance, responsive interface focused on accessibility (a11y) and intuitive UX/UI for diverse patient demographics.
- Integrated Firebase Authentication and Firestore for secure data handling and real-time scheduling updates.